

Product Requirements Document

AI Multi-Agent Candidate Evaluation & Hiring System

Version: 1.1

Status: Revised Draft for Technical Design

Product Type: AI-powered recruitment intelligence platform

Primary Users: Recruiters, hiring managers, interviewers

Target: Student hackathon / academic project MVP

1. Executive Summary

The AI Multi-Agent Candidate Evaluation & Hiring System is an evidence-driven recruitment platform that evaluates candidates using their resume, interview transcript, and target job description.

Instead of relying on a single AI model, the system uses four specialized AI personas that independently evaluate the candidate from different perspectives. Their independent conclusions are then exposed to a controlled debate, after which a separate Final Decision Agent reasons over the evidence, disagreements, confidence, role requirements, and debate outcomes.

The system consists of:

1. A neutral Candidate Profile Builder that extracts and normalizes facts, claims, and evidence without making hiring judgments.
2. Four independent AI evaluators: Technical Agent, HR/Culture Agent, Hiring Manager Agent, and Skeptic Agent.
3. A controlled multi-agent debate where agents challenge and respond to one another.
4. A separate Final Decision Agent that makes a reasoning-based recommendation rather than averaging scores.
5. An interactive Final Candidate Report with full traceability from decision to source evidence.

Core principle: Evidence first. Independent judgment second. Debate third. Decision last.

2. Problem Statement

Traditional candidate screening can suffer from over-reliance on resumes, inconsistent evaluation, difficulty comparing technical and behavioral evidence, unverified claims, confirmation bias, contradictions, lack of transparency, and recommendations that provide scores without explaining why.

A single AI evaluator can also inherit the limitations of a single perspective.

This system addresses the problem by creating multiple independent evaluators, forcing them to justify assessments using evidence, allowing them to challenge one another, and then using a separate reasoning stage to reach a final recommendation.

3. Product Vision

Create a recruitment intelligence system where a hiring decision can be explored as an evidence-backed reasoning process rather than a black-box AI score.

The recruiter should be able to answer:

- What does the candidate claim?
- What evidence supports that claim?
- What does each evaluator think independently?
- Where do the evaluators disagree?
- Did the debate change anyone's opinion?

- Why did the system ultimately recommend this candidate?

4. Goals

Primary Goals

- Accept a job description.
- Accept a candidate resume.
- Accept an interview transcript.
- Extract a neutral, structured Candidate Profile.
- Preserve evidence and source references.
- Run at least four independent AI evaluations.
- Make independent evaluation boundaries architecturally enforceable and observable.
- Require evidence-backed findings using real source quotes or structured facts.
- Validate agent-provided evidence against source documents.
- Conduct a genuine multi-agent debate.
- Require substantive responses to previous arguments.
- Allow agents to challenge and revise opinions.
- Track opinion changes.
- Use a separate final reasoning stage.
- Prevent arithmetic aggregation from determining the recommendation.
- Map job requirements to candidate evidence.
- Generate an interactive final report.
- Make final conclusions fully traceable to source evidence.

Secondary Goals

- Make the evaluation understandable to non-technical users.
- Provide useful progress indicators during long-running AI operations.
- Make disagreements easy to identify.
- Make uncertainty explicit.
- Provide a strong visual demonstration of multi-agent reasoning.

5. Non-Goals

The MVP will NOT attempt to:

- Fully automate hiring without human oversight.
- Make legally binding employment decisions.
- Guarantee objective candidate suitability.
- Replace professional recruiters.
- Perform unrestricted autonomous agent conversations.
- Build a large-scale enterprise ATS.
- Support every possible document format.

- Implement complex multi-agent infrastructure unnecessarily.
- Depend on voice interaction for the core product.

Voice debate is an optional enhancement and must not be required for the MVP.

6. Target Users

Recruiter

Needs to upload candidate information, quickly understand strengths and concerns, inspect supporting evidence, and understand the recommendation.

Hiring Manager

Needs to evaluate candidate-job fit, understand technical and practical suitability, investigate disagreements, and make an informed decision.

Interviewer / Evaluator

Needs to review interview evidence, compare claims against demonstrated knowledge, and identify areas requiring further verification.

7. Core User Journey

Create Evaluation → Add Job Description → Upload Resume → Upload Interview Transcript → Document Processing → Neutral Candidate Profile Builder → Recruiter Reviews Profile → Independent Agent Evaluations → All 4 Evaluations Complete → Multi-Agent Debate → Agents Respond / Challenge / Revise → Final Decision Agent → Final Candidate Report → Evidence Exploration

8. Functional Requirements

FR-01 — Create Evaluation

The user must be able to create an evaluation for a target role.

Inputs:

- Job title
- Job description

Output:

A unique evaluation containing the target role and job requirements.

Acceptance criteria:

- User can create an evaluation.
- Job description is stored.
- Evaluation receives a unique identifier.

9. Document Upload

The system must allow the user to provide:

- Resume
- Interview/transcript
- Job description

Supported MVP formats:

- PDF
- DOCX
- TXT
- Pasted text

Requirements:

- Validate file type.
- Apply file-size limits.
- Reject unsupported files.
- Extract text.
- Preserve source metadata where possible.

For PDFs preserve page, section, and paragraph where available. For transcripts preserve timestamp, speaker, and section where available.

10. Candidate Profile Builder

The Candidate Profile Builder is the shared information layer for all agents.

It receives:

Resume + Interview Transcript + Job Description

and produces a structured Candidate Profile.

10.1 Neutrality Requirement

The Candidate Profile Builder must extract, normalize, classify, and link information to evidence.

It must NOT:

- Recommend hire/reject.
- Score the candidate.
- Judge candidate quality.
- Assign personality traits.
- Resolve disputed claims in favor of the candidate.
- Generate agent-style conclusions.

This prevents an early AI judgment from influencing all downstream agents.

10.2 Extracted Information

Where available:

- Candidate name
- Education
- Skills
- Technical skills
- Programming languages
- Tools
- Frameworks
- Projects

- Work experience
- Certifications
- Achievements
- Quantifiable achievements
- Candidate claims
- Communication evidence
- Teamwork evidence
- Leadership evidence
- Potential inconsistencies
- Evidence supporting extracted information

10.3 Information Classification

FACT: Directly supported by the source.

CANDIDATE CLAIM: Something the candidate states but which may require verification.

AGENT INTERPRETATION: A conclusion made by an evaluator. This category must not be generated by the Profile Builder.

Interpretations must never be silently stored as facts.

11. Evidence System

Evidence is a first-class product object.

Every evidence item must have a unique ID and, where available:

- Source document ID
- Source type
- Page / paragraph / section or timestamp
- Speaker for transcripts
- Exact quote or extracted fact
- Context
- Evidence type
- Extraction confidence

11.1 Evidence Types

DIRECT_QUOTE: Exact text from a source document.

STRUCTURED_FACT: A normalized fact directly supported by source text.

INFERENCE: An agent interpretation based on one or more evidence items. Inferences are not source facts.

UNSUPPORTED: A proposed finding for which sufficient source evidence cannot be established.

11.2 Evidence Rules

Every important agent finding must reference one or more Evidence IDs.

A finding may only be presented as evidence-backed if its Evidence IDs resolve to source material.

Agents must never invent quotations.

If sufficient evidence cannot be found, the agent must state:

Insufficient evidence.

11.3 Evidence Validation

Agent-provided quotes must be validated against the stored source text before they are accepted as valid evidence.

Validation flow:

Agent finding → Evidence ID / quote → Source lookup → Exact or validated match → Accept / Reject

If a quote cannot be validated:

- Do not display it as verified evidence.
- Attempt a controlled agent retry if configured.
- Otherwise mark the finding as unsupported or insufficiently evidenced.

The system must never silently convert an invalid quote into verified evidence.

12. Job Requirement Extraction & Mapping

The Job Description must be converted into structured role requirements.

Each requirement should contain:

- Requirement ID
- Requirement text
- Category
- Priority: Must Have / Preferred / Optional
- Evidence needed to demonstrate it

Candidate evidence should then be mapped to requirements.

Example:

Requirement R1: Python — Must Have

Evidence E14, E22 → Supported

Requirement R2: 3+ years backend experience — Must Have

Evidence E31 → Partially Supported

Requirement R3: Team collaboration — Preferred

Evidence E42 → Supported

The mapping must distinguish:

- Supported
- Partially Supported
- Not Supported
- Unknown / Insufficient Evidence

The requirement mapping must not itself make the final hiring decision.

13. AI Agent Architecture

The MVP contains four independent evaluator personas:

1. Technical Agent
2. HR/Culture Agent
3. Hiring Manager Agent

4. Skeptic Agent

All four receive the same neutral Candidate Profile and role requirements, but each uses its own persona instructions and evaluation criteria.

14. Independent Evaluation Boundary

This is a critical system requirement.

Before the debate begins:

No agent may access another agent's independent conclusion.

Each independent evaluation must be performed through a separate LLM call.

14.1 Allowed Input

An independent agent may receive:

- Candidate Profile
- Job Requirements
- Source-linked Evidence
- Its own system instructions
- Its own evaluation rubric

14.2 Forbidden Input

An independent agent must not receive:

- Another agent's score
- Another agent's findings
- Another agent's reasoning
- Another agent's recommendation
- Debate messages
- Final decision output
- Any hidden aggregate of other agents

14.3 Architectural Enforcement

Independent calls must use isolated input objects derived only from approved shared inputs.

The output of one independent call must not be inserted into the input of another independent call.

There must be no shared opinion state between independent calls.

The orchestrator must record execution-stage metadata showing:

- Agent
- Stage
- Allowed input sources
- Excluded agent outputs
- Evaluation completion status

This metadata is for auditability and must not expose sensitive information unnecessarily.

15. Technical Agent

Purpose: Evaluate technical capability and depth.

Evaluation areas:

- Programming languages
- Technical skills
- Frameworks
- Tools
- Projects
- Problem solving
- Technical depth
- Practical experience
- Relevant experience
- Technical interview answers
- Surface-level versus demonstrated expertise

The agent must distinguish claimed skills from demonstrated skills and identify missing evidence.

Output:

- Overall assessment
- Findings
- Strengths
- Concerns
- Evidence IDs
- Confidence
- Technical recommendation
- Uncertainties

16. HR / Culture Agent

Purpose: Evaluate behavioral and interpersonal suitability using role-relevant evidence.

Evaluation areas:

- Communication
- Teamwork
- Collaboration
- Leadership
- Ownership
- Adaptability
- Professionalism
- Transparency
- Behavioral interview evidence

The agent must not infer personality from irrelevant demographic information.

Every significant conclusion must be supported by evidence or explicitly marked as uncertain.

17. Hiring Manager Agent

Purpose: Evaluate whether the candidate is suitable for the specific target role.

Consider:

- Job requirements
- Candidate-job fit
- Relevant experience
- Technical capability
- Practical value
- Potential
- Strengths
- Risks
- Missing requirements
- Requirement coverage

The agent should reason as:

“Would I recommend this candidate for this role, given the available evidence?”

18. Skeptic Agent

Purpose: Act as an adversarial evaluator.

Search for:

- Contradictions
- Unsupported claims
- Exaggeration
- Resume/transcript inconsistencies
- Vague answers
- Inflated achievements
- Missing evidence
- Potential red flags
- Overstated technical expertise

Concerns must be classified as:

- Proven Concern
- Plausible Concern
- Insufficient Evidence
- No Concern

The Skeptic must not assume a claim is dishonest merely because it lacks evidence.

19. Independent Evaluation Output

Every agent must produce structured output containing:

- Agent
- Evaluation ID
- Overall assessment
- Recommendation

- Confidence
- Findings
- Evidence IDs
- Strengths
- Concerns
- Uncertainties
- Requirement mappings where relevant

Confidence must be defined as:

High: Evidence is direct, relevant, and consistent; major uncertainties are limited.

Medium: Evidence supports the conclusion but meaningful uncertainty remains.

Low: Evidence is sparse, contradictory, indirect, or highly uncertain.

Every confidence level must include a short justification.

20. Debate Engine

The debate is a core product feature. Simply displaying four independent opinions is insufficient.

The debate receives:

- Candidate Profile
- Job Requirements
- Four independent evaluations
- Evidence references

During debate, agents may see the other agents' independent evaluations.

Agents may:

- Agree
- Disagree
- Challenge another agent
- Defend a conclusion
- Identify missing evidence
- Reinterpret evidence
- Correct an interpretation
- Revise confidence
- Revise recommendation

20.1 Debate Protocol

Round 1 — Challenge: Identify substantive disagreements.

Round 2 — Response: Respond to opposing arguments and evidence.

Round 3 — Revision: State whether the assessment changed and why.

The debate must terminate after a configured maximum number of rounds.

20.2 Substantive Debate Requirement

A valid debate response must:

- Reference a specific prior argument or message.
- Address the substance of that argument.
- Use evidence where appropriate.
- Clearly agree, disagree, qualify, or correct.

A message such as “I agree” without substantive reasoning does not count as debate.

At least one agent must directly respond to another agent's point.

21. Debate Message Model

Each debate message should contain:

- Message ID
- Debate round
- Agent
- Statement
- Responding-to message ID
- Position: Agree / Disagree / Qualify / Correct
- Evidence IDs
- Confidence
- Timestamp

This allows the UI and backend to prove that a real response occurred.

22. Opinion Revision

Agents must be able to change their position.

The system must record:

- Original position
- New position
- Original confidence
- New confidence
- What caused the change
- Debate message(s) that caused it
- Evidence involved

Example:

Technical Agent: High confidence → Medium confidence

Reason: Skeptic identified that production-scale experience was not sufficiently demonstrated.

23. Final Decision Agent

The final decision is a separate reasoning stage.

It receives:

- Candidate Profile
- Job Requirements
- Independent Evaluations

- Evidence
- Confidence levels
- Debate messages
- Revised opinions
- Requirement mappings
- Unresolved disagreements

23.1 Non-Aggregation Requirement

The final recommendation must NOT be determined by:

- Simple average
- Weighted average
- Majority vote alone
- Highest agent score
- A hidden arithmetic aggregate

The system must not provide a precomputed aggregate score as the primary decision input.

The Final Decision Agent must reason over evidence and competing interpretations.

23.2 Decision Factors

Consider:

- Evidence quality
- Evidence relevance
- Requirement coverage
- Confidence
- Concern severity
- Agent agreement
- Contradictions
- Debate outcomes
- Opinion revisions
- Uncertainty

23.3 Decision Output

The final output must contain:

- Recommendation
- Confidence
- Decision rationale
- Key supporting evidence IDs
- Key concerns
- Resolved disagreements
- Unresolved disagreements
- Decision factors

- Why the recommendation was selected over plausible alternatives

The final reasoning must be traceable to underlying findings and evidence.

24. Final Recommendation

Recommendation options:

- Strong Hire
- Hire
- Consider
- Weak Consider
- Reject

The recommendation must include:

- Reasoning
- Confidence
- Supporting evidence
- Concerns
- Unresolved disagreements
- Explanation of why stronger or weaker alternatives were not selected

25. Final Report

The report is the main user-facing output.

It should include:

- Candidate summary
- Target role
- Final recommendation
- Confidence
- Decision explanation
- Evidence-backed strengths
- Evidence-backed concerns
- Requirement coverage
- Agent perspectives
- Debate summary
- Position changes
- Unresolved disagreements
- Interactive evidence explorer
- Full reasoning trace

25.1 Traceability Chain

Every important final conclusion should be traceable through:

Final Decision → Decision Factor → Agent Finding → Evidence ID → Source Document → Exact Quote / Source Fact

If a conclusion cannot be traced to sufficient evidence, it must be labeled as uncertain or unsupported.

26. UI/UX Product Direction

The product must NOT look like a generic AI SaaS dashboard.

It should feel like a professional recruitment intelligence and evidence-analysis platform.

Avoid:

- Excessive purple gradients
- Glowing AI effects
- Robot illustrations
- Generic chatbot UI
- Giant AI labels
- Decorative dashboards
- Excessive rounded cards
- Meaningless animations

Desired:

- Professional
- Analytical
- Trustworthy
- Evidence-driven
- Modern
- Interactive
- Clear
- Human-centered

The interface should visually communicate:

Evidence → Analysis → Disagreement → Debate → Decision

27. Interactive Experience Requirements

Interactivity is a functional requirement, not only a visual preference.

Users must be able to:

- Click evidence to see the findings that reference it.
- Click a finding to see supporting source material.
- Click an agent conclusion to see where it was challenged.
- Jump from debate messages to evidence.
- Inspect original versus revised positions.
- Filter debate by agent.
- Navigate between debate rounds.
- Trace final decision factors back to evidence.

28. Core UI Screens

Dashboard

Shows:

- Candidates
- Target roles
- Evaluation status
- Progress
- Recent evaluations

Create Evaluation

User selects/creates role and uploads job description, resume, and transcript.

Candidate Profile

Interactive sections:

- Overview
- Experience
- Skills
- Projects
- Claims
- Requirements
- Evidence

Agent Analysis

Users switch between Technical, HR/Culture, Hiring Manager, and Skeptic.

Display:

- Findings
- Evidence
- Confidence
- Strengths
- Concerns
- Uncertainties
- Requirement mappings

29. Debate Workspace

The Debate Workspace is a major differentiator.

It should feel like a structured professional discussion, not a chatbot.

Each message should show:

- Agent
- Statement
- Responding-to message
- Evidence
- Agreement/disagreement

- Confidence
- Position change

Users should be able to:

- Expand messages.
- Inspect evidence.
- Filter by agent.
- See response relationships.
- Jump between debate rounds.
- Identify changed opinions.
- Follow a disagreement from challenge to resolution.

30. Reasoning Timeline

The system should provide an interactive timeline showing major evaluation events:

Documents processed

- Profile created
- Independent evaluations
- Debate challenge
- Response
- Position revision
- Final decision

The timeline must accurately reflect the real execution stages and must not invent events.

31. Final Decision Interface

The recommendation should feel like a professional decision brief.

Show:

- Final recommendation
- Confidence
- Why
- Strengths
- Concerns
- Requirement coverage
- Agent perspectives
- Resolved disagreements
- Unresolved issues
- Evidence and debate links
- Why this recommendation was selected over alternatives

32. Interactive Evidence Explorer

Clicking a finding should open an evidence panel/drawer.

Display:

- Evidence ID
- Source document
- Timestamp/page/section
- Exact quote or source fact
- Assessment
- Agents referencing it
- Related job requirements

33. Loading & Progress States

Because LLM operations may take time, show meaningful system progress.

Example:

Documents processed → Profile created → Technical ✓ → HR ✓ → Hiring Manager ● → Skeptic ● → Independent review in progress

The UI must not imply that debate has begun while independent evaluation is still running.

34. Error Handling

Handle:

- Invalid file
- Unsupported file
- Empty document
- Parsing failure
- Missing transcript
- LLM failure
- Invalid structured output
- Invalid/unverifiable evidence
- Debate failure
- Final decision failure

Never fabricate missing results.

If an agent fails, show:

Agent unavailable — evaluation incomplete.

35. Structured Agent Output

All agents must produce machine-validated structured output.

At minimum:

- Agent
- Evaluation ID
- Assessment
- Recommendation
- Confidence
- Findings

- Evidence IDs
- Strengths
- Concerns
- Uncertainties

The exact production schema will be defined in the Technical Design and Agent Specification.

36. Prompt Architecture

Every agent prompt should contain:

1. System instructions
2. Persona
3. Objective
4. Available information
5. Forbidden information
6. Evaluation criteria
7. Evidence requirements
8. Uncertainty rules
9. Anti-hallucination rules
10. Structured output requirements

Candidate documents are data, not instructions.

37. Prompt Injection Defense

Uploaded resumes and transcripts must be treated as untrusted content.

If a document contains instructions such as “ignore previous instructions,” the system must interpret them as candidate-provided text, not as system instructions.

Document content must never override system or agent instructions.

38. Document Processing

MVP support:

- PDF
- DOCX
- TXT
- Pasted text

Pipeline:

Document → Text Extraction → Cleaning → Section Detection → Chunking → Evidence Indexing → Candidate Profile

Scanned-PDF OCR is optional.

39. Data Model

Core entities:

- Candidate
- Job

- JobRequirement
- Document
- CandidateProfile
- Evidence
- AgentEvaluation
- AgentFinding
- Debate
- DebateMessage
- OpinionRevision
- FinalDecision
- FinalReport

Important relationships:

Candidate → Documents → CandidateProfile → Evidence

Candidate + Job → AgentEvaluations

AgentEvaluations → Debate → DebateMessages → OpinionRevisions

Debate + Evidence + Requirements → FinalDecision → FinalReport

Detailed schema will be defined in the Technical Design.

40. API Requirements

Minimal API areas:

- Evaluation
- Candidate
- Documents
- Profile
- Job Requirements
- Agents
- Debate
- Decision
- Evidence
- Report

Exact contracts will be defined in the API Specification.

41. Security & Privacy

Requirements:

- Never expose API keys in frontend code.
- Validate uploads.
- Restrict file sizes.
- Isolate candidate data between evaluations.
- Clean temporary files.
- Restrict access where authentication exists.

- Avoid unnecessary exposure of sensitive information.
- Avoid unnecessary sensitive candidate data in logs.
- Prevent cross-candidate data leakage.

42. Fairness

The system is a decision-support tool.

The MVP should:

- Avoid protected attributes as evaluation criteria.
- Avoid demographic assumptions.
- Avoid inferring personality from irrelevant information.
- Base findings on role-relevant evidence.
- Make uncertainty visible.
- Allow human review.

43. Observability & Auditability

Track enough information to demonstrate and debug:

- LLM calls
- Agent execution status
- Evaluation stage
- Latency
- Token usage where available
- Validation failures
- Evidence validation status
- Debate rounds
- Message response relationships
- Opinion revisions
- Final decision status

The system should retain an auditable record of:

- What each independent agent was allowed to see.
- What each independent agent produced.
- When independence ended.
- What was shown during debate.
- Which arguments caused revisions.
- Which evidence supported the final decision.

Sensitive candidate content should not be unnecessarily duplicated in logs.

44. Performance & Cost

Recommended flow:

Profile Builder → 4 independent LLM calls → 1–3 controlled debate rounds → 1 final decision call

The four independent evaluations can run concurrently.

Avoid repeatedly sending entire documents when structured evidence can be referenced instead.

Evidence validation may require additional controlled processing and should be optimized without weakening traceability.

45. Failure Recovery

If an agent fails:

- Do not fabricate its evaluation.
- Mark it unavailable.
- Allow configured retry.
- Record failure status.

If structured output is invalid:

- Validate.
- Retry in a controlled manner.
- Reject persistent invalid output.

If evidence is missing or unverifiable:

- Mark the finding unsupported or insufficiently evidenced.

If final decision generation fails:

- Do not display a fabricated recommendation.

46. MVP Scope

MUST BUILD:

- Job description, resume, transcript input
- Document parsing
- Neutral Candidate Profile Builder
- Evidence extraction and validation
- Job requirement extraction and evidence mapping
- Four agents
- Strict independent execution boundary
- Auditable independence metadata
- Debate
- Substantive direct response
- Opinion revision
- Final reasoning
- Non-aggregated final recommendation
- Confidence
- Strengths
- Concerns
- Agent perspectives
- Debate summary
- Unresolved disagreements
- Full evidence traceability

- Dashboard
- Upload/evaluation screen
- Candidate Profile
- Agent Analysis
- Debate Workspace
- Reasoning Timeline
- Final Decision
- Evidence Explorer

47. Future Features

Potential V2:

- Voice debate
- Candidate comparison
- Multiple job roles
- Recruiter feedback
- Human override
- Interview question generation
- Historical candidate analytics
- Advanced RAG
- Candidate ranking
- Recruiter collaboration
- Additional specialist agents

Voice debate is a bonus feature and must not be required for MVP.

48. Acceptance Criteria

Candidate Profile

- Resume processed.
- Transcript processed.
- Structured profile created.
- Profile contains facts and claims.
- Profile does not make hiring judgments.
- Evidence stored.

Job Requirements

- Job requirements extracted.
- Requirements have priority.
- Candidate evidence can be mapped to requirements.
- Mapping supports Supported / Partially Supported / Not Supported / Unknown.

Independent Agents

- Four personas exist.
- Separate LLM calls.
- No cross-agent conclusions before debate.
- Independent evaluations complete before debate.
- Backend can demonstrate allowed/excluded input sources.
- No shared opinion state exists between independent calls.

Evidence

- Important findings contain evidence.
- Evidence contains source information.
- Quotes are traceable.
- Agent quotes are validated.
- Invalid quotes are rejected or marked unsupported.
- No fabricated evidence.
- Insufficient evidence explicitly reported.

Debate

- Debate occurs after independent evaluation.
- Agents see other evaluations only during debate.
- At least one direct response.
- Responses reference specific prior messages.
- Responses are substantive.
- Agents can disagree.
- Agents can defend positions.
- Agents can revise positions.
- Confidence can change.
- Debate terminates in a controlled way.

Final Decision

- Separate reasoning stage.
- Uses evidence.
- Considers disagreements.
- Considers confidence.
- Considers job requirements.
- Considers debate outcomes.
- Not simple score averaging.
- Does not rely on majority vote alone.
- Explains why the chosen recommendation was preferred over alternatives.
- Every major conclusion is traceable.

Final Report

- Recommendation.
- Confidence.
- Strengths.
- Concerns.
- Requirement coverage.
- Agent perspectives.
- Debate summary.
- Opinion revisions.
- Unresolved disagreements.
- Evidence traceability.
- Interactive exploration.

49. Critical Demo Scenario

The demonstration candidate should contain mixed evidence:

- Strong technical skills.
- Impressive project claims.
- A questionable claim.
- A subtle resume/transcript inconsistency.
- Behavioral evidence.
- At least one meaningful disagreement.

Expected flow:

Candidate → Profile → Four independent opinions → Debate → Direct challenge/response → Opinion revision → Final decision → Interactive report

The final decision should be nuanced, for example:

HIRE — Medium/High confidence, with strong relevant technical evidence but partially unverified production-scale experience.

The demo should visibly prove:

1. Agents first formed independent opinions.
2. One agent challenged another using evidence.
3. Another agent responded.
4. At least one agent changed or qualified its position.
5. The final decision considered the disagreement rather than averaging scores.
6. The final recommendation can be traced back to source evidence.

50. Risks & Mitigations

- LLM hallucinated evidence → Evidence IDs + source validation.
- Agents influence each other too early → Isolated independent execution.
- Independence is difficult to prove → Execution metadata + explicit stage boundaries.
- Debate becomes fake → Response IDs + substantive-response validation.

- Endless debate → 1–3 controlled rounds.
- Final decision becomes score average → Explicit non-aggregation requirement.
- Candidate Profile becomes hidden evaluator → Neutral profile-builder contract.
- Job fit is vague → Explicit requirement extraction and evidence mapping.
- Prompt injection → Treat documents as untrusted data.
- Missing evidence → Explicit uncertainty.
- Agent failure → Retry / graceful degradation.
- Generic AI UI → Evidence-first recruitment UX.
- Excessive complexity → Simple orchestrator + modular services.
- Bias → Role-relevant evidence and human oversight.
- High API cost → Parallel calls + structured evidence.

51. Recommended Architecture

Frontend → Backend API → Document Processing + Profile Builder → Evidence Store → Independent Agent Service → Debate Engine → Final Decision Agent → Report / Evidence Explorer

Critical stage boundaries:

Stage 1 — Neutral Processing

Documents → Profile + Evidence + Requirements

Stage 2 — Independent Evaluation

Profile + Evidence + Requirements → Four isolated LLM calls

Stage 3 — Debate

Independent outputs → Controlled multi-agent interaction

Stage 4 — Final Decision

All evidence + evaluations + debate → Separate reasoning call

Stage 5 — Report

Decision + traceability graph → Interactive report

The architecture must make the Stage 2 independence boundary explicit and auditable.

52. Recommended Technology Direction

Frontend: React / Next.js

Backend: Python + FastAPI

Database: PostgreSQL

LLM: One capable provider with structured output support

Document Processing: Python PDF/DOCX parsing libraries

Storage: Local development storage; managed object storage if required

Agent Orchestration: Lightweight custom Python orchestration preferred for MVP

Prioritize transparent orchestration over framework complexity.

53. Product Design Principles

1. Evidence over scores.

2. Independent reasoning before debate.
3. Debate over parallel summaries.
4. Reasoning over arithmetic.
5. Transparency over black box.
6. Uncertainty over hallucination.
7. Traceability over unsupported claims.
8. Interaction over static display.
9. Professional recruitment UX over generic AI aesthetics.
10. Human oversight over autonomous hiring.

54. End-to-End Requirement Trace

Resume + Transcript + Job Description

- Document Parser
- Neutral Candidate Profile + Evidence + Job Requirements
- Four Independent Agent Calls
- Independent Outputs
- Debate
- Direct Responses
- Opinion Revision
- Final Decision Agent
- Evidence-weighted Recommendation
- Traceability Graph
- Interactive Final Report

55. Definition of Done

The project is ready for demonstration when:

1. Recruiter can create an evaluation.
2. Resume, transcript, and job description can be supplied.
3. Candidate Profile Builder produces structured, neutral information.
4. Evidence is traceable and quote validation works.
5. Job requirements are extracted and mapped to evidence.
6. Four agents independently evaluate the candidate.
7. The application can demonstrate that agents did not see one another's conclusions before debate.
8. Independent input boundaries are auditable.
9. A controlled debate occurs.
10. At least one agent directly responds substantively to another.
11. At least one agent can revise or qualify its opinion.
12. Revision records identify what caused the change.
13. Final Decision Agent produces a reasoning-based recommendation.
14. Recommendation is not calculated by averaging or majority vote alone.

15. Final report contains all required information.
16. Users can explore supporting evidence.
17. Final conclusions have a traceability chain.
18. Prompt injection inside candidate documents does not override agent instructions.
19. Failure states are handled gracefully.
20. Interface feels like a professional recruitment intelligence platform rather than a generic AI dashboard.

56. Success Metric for the Hackathon

The strongest demonstration is not simply:

“We have four AI agents.”

It is:

Four independent evaluators examined the same evidence without influencing each other. They then challenged one another, changed or qualified their positions where appropriate, and a separate reasoning stage explained why the final hiring recommendation was reached.

The product should make this process visible, interactive, auditable, and traceable to the source documents.